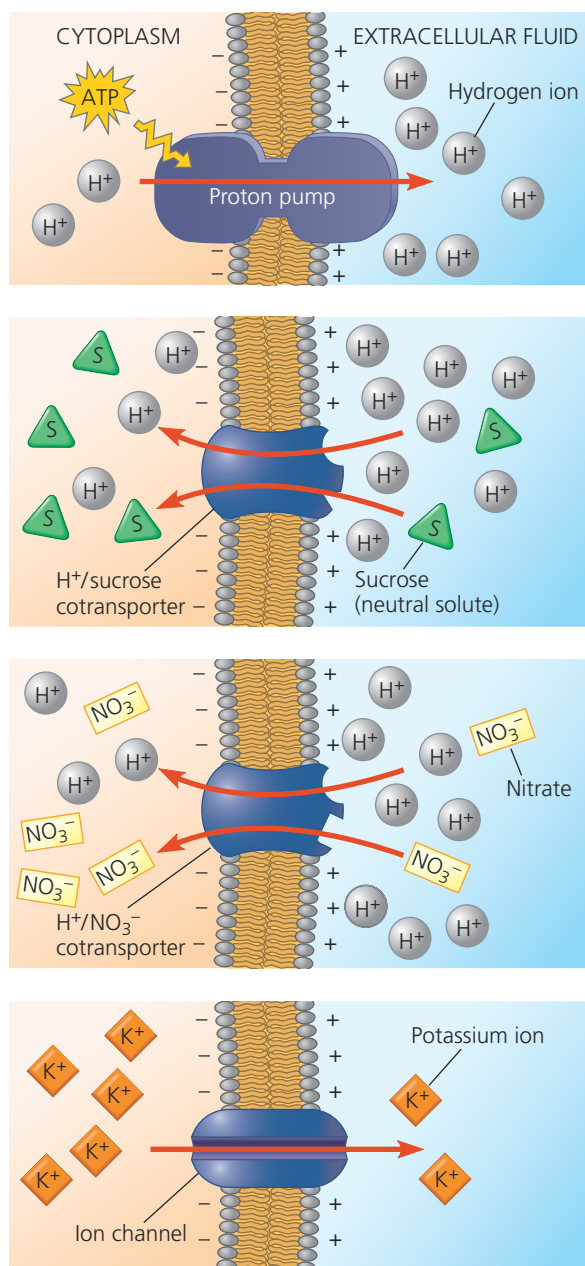


▼ **Figure 36.6** Solute transport across plant cell plasma membranes.



(a) H⁺ and membrane potential.

The plasma membranes of plant cells use ATP-dependent proton pumps to pump H⁺ out of the cell. These pumps contribute to the membrane potential and the establishment of a pH gradient across the membrane. These two forms of potential energy can drive the transport of solutes.

(b) H⁺ and cotransport of neutral solutes.

Neutral solutes such as sugars can be loaded into plant cells by cotransport with H⁺ ions. H⁺/sucrose cotransporters, for example, play a key role in loading sugar into the phloem prior to sugar transport throughout the plant.

(c) H⁺ and cotransport of ions.

Cotransport mechanisms involving H⁺ also participate in regulating ion fluxes into and out of cells. For example, H⁺/NO₃⁻ cotransporters in the plasma membranes of root cells are important for the uptake of NO₃⁻ by plant roots.

(d) Ion channels.

Plant ion channels open and close in response to voltage, stretching of the membrane, and chemical factors. When open, ion channels allow specific ions to diffuse across membranes. For example, a K⁺ ion channel is involved in the release of K⁺ from guard cells when stomata close.

? Assume that a plant cell has all four of the plasma membrane transport proteins shown above and that you have a specific inhibitor for each protein. Predict the effect of each inhibitor on the cell's membrane potential.

examined in the **Scientific Skills Exercise**, which explores the effect of temperature on this process.

Water potential is abbreviated by the Greek letter Ψ (psi, pronounced “sigh”). Plant biologists measure Ψ in a unit of pressure called a **megapascal** (abbreviated MPa). By definition, the Ψ of pure water in a container open to the atmosphere under standard conditions (at sea level and at room temperature) is 0 MPa. One MPa is equal to about 10 times atmospheric pressure at sea level. The internal pressure of a living plant cell due to the osmotic uptake of water is approximately 0.5 MPa, about twice the air pressure inside an inflated car tire.

How Solutes and Pressure Affect Water Potential

Solute concentration and physical pressure are the major determinants of water potential in hydrated plants, as expressed in the water potential equation:

$$\Psi = \Psi_s + \Psi_p$$

where Ψ is the water potential, Ψ_s is the solute potential (osmotic potential), and Ψ_p is the pressure potential. The **solute potential** (Ψ_s) of a solution is directly proportional to its molarity. Solute potential is also called *osmotic potential* because solutes affect the direction of osmosis. The solutes in plants are typically mineral ions and sugars. By definition, the Ψ_s of pure water is 0. When solutes are added, they bind water molecules. As a result, there are fewer free water molecules, reducing the capacity of the water to move and do work. In this way, an increase in solute concentration has a negative effect on water potential, which is why the Ψ_s of a solution is always expressed as a negative number. For example, a 0.1 M solution of a sugar has a Ψ_s of -0.23 MPa. As the solute concentration increases, Ψ_s will become more negative.

Pressure potential (Ψ_p) is the physical pressure on a solution. Unlike Ψ_s , Ψ_p can be positive or negative relative to atmospheric pressure. For example, when a solution is being withdrawn by a syringe, it is under negative pressure; when it is being expelled from a syringe, it is under positive pressure. The water in living cells is usually under positive

pressure due to the osmotic uptake of water. Specifically, the **protoplast** (the living part of the cell, which also includes the plasma membrane) presses against the cell wall, creating what is known as **turgor pressure**. This pushing effect of internal pressure, much like the air in an inflated tire, is critical for plant function because it helps maintain the stiffness of plant tissues and also serves as the driving force for cell elongation. Conversely, the water in the hollow nonliving xylem cells (tracheids and vessel elements) of a plant is often under a negative pressure potential (tension) of less than -2 MPa.